

Traveling Wave Protection for the AC Transmission Lines between MMC Stations and Renewable Energy Sources

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Abstract—Several MMC-HVDC systems with renewable energy sources connected have been constructed in the past decades. For the AC transmission lines between the MMC stations and the renewable energy sources, fault-ride-through strategies at both ends will lead to complicated fault characteristics. Specifically, the current amplitudes are restricted and the phase angles are controlled, the characteristics decrease the reliability and sensitivity of traditional relay protection. This paper proposes a differential protection based on the traveling wave theory, which focuses on the transmission lines and excludes the influence of the power sources outside the protected area. Criterion is constructed based on the integral value of the amplitude of the traveling wave differential current. The proposed protection is verified in a PSCAD simulation model. Test results prove that, compared with traditional current differential protection, the traveling wave differential protection has advantages in reliability, and the ability to withstand fault resistance.

Keywords—MMC-HVDC; renewable energy; traveling waves; traveling wave differential protection

I. INTRODUCTION

As an option for mitigating environmental problems, renewable energy sources (RES) such as photovoltaic and wind power are developing rapidly and DC transmission technology is a practical solution for connecting RES to the power systems. High-voltage DC technology based on modular multilevel converters (MMC-HVDC) has advantages of high efficiency, high modularity, high scalability, and few harmonics[1], it has become an effective and practical solution for RES integration and has been widely constructed, for instance, the Zhangbei DC grid in China[2]. However, new requirements for protection have emerged such as fast operation ability on the DC side[3], it is also necessary to deploy suitable protection on the AC side[4].

Transmission lines at the AC side of an MMC-HVDC system are lines with MMC stations at one end and RES at the other, as shown in Fig. 1. RES includes permanent magnet synchronous generator (PMSG), doubly-fed induction generator (DFIG), and photovoltaic (PV).

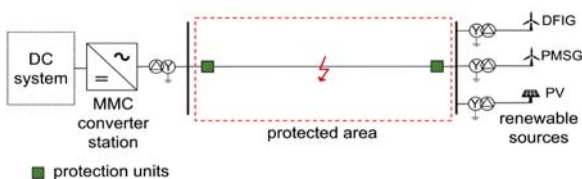


Fig. 1. AC transmission line between MMC station and RES.

MMC station and RES will adopt coordinated control strategies when a fault occurs, leading to complicated fault characteristics. In addition to the restricted amplitudes[5], phase angles of the fault current are controlled. These characteristics will bring challenges for traditional relay protections based on the fundamental frequency phasor measurement[4] such as current differential protection and distance protection.

The widely used current differential protection is based on the algebraic summation of the current flowing into and out of the protected area to identify internal faults[6]. The restricted fault current may lead to a restricted differential current. Moreover, a restraining current is needed to prevent the influence of the unbalanced current. When the phase angles are controlled, the differential current may become smaller than the restraining current, the protection may fail to trip. As fault resistance increases, fault features become harder to detect[7], this effect will be worsened considering the fault characteristics of the system.

Therefore, it is of great significance to propose a highly sensible protection scheme. The scheme should be able to exclude the influence of power sources out of the protected area and withstand higher fault resistance.

Traveling wave (TW) contains a wealth of fault information. Protections based on the traveling waves are immune to distributed capacitance current, fault resistance, and power system swing[8]. Among different types of traveling wave protection, differential protection fully utilizes information in traveling waves[9], and can reliably operate with high speed. In [10,11], traveling wave differential protection was firstly discussed, so far, many traveling wave differential protection methods have been proposed, most of them use the first wavefronts, placing higher demands on sampling, communication and timing calibration technology.

In this paper, traveling wave differential protection is applied to the AC transmission lines between MMC stations and RES. In Section II, time-domain expressions of current at both ends are briefly deduced to describe the fault characteristics. Section III introduces the traveling wave differential protection scheme applicable to the system. In Section IV, we verify the feasibility and performance of the proposed protection in the simulation model. The paper is then concluded in Section V.

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II. FAULT CHARACTERISTICS OF THE SYSTEM

A. Characteristics of Power Sources at Both Ends

After a fault, inverter-interfaced renewable energy generation (IIREG, including PMSGs and PVs) usually adopt a control strategy to suppress the negative-sequence current[12]. MMC station restricts the current by generating a reduced reference voltage according to the virtual frequency. In the faulted network, IIREG and MMC station can be modeled as a positive-sequence current source and a positive-sequence voltage source respectively.

As for DFIGs, crowbar protection may be engaged depending on the rotor current, and the current will decay to a steady-state value within tens of milliseconds. The voltage loop is cut off after the fault and DFIGs can be modeled as current sources. Negative-sequence current is indispensable to avoid the increased AC voltage[13], so the DFIGs supply both positive and negative-sequence currents.

Characteristics of power sources at both ends can be modeled as TABLE I.

TABLE I. FAULT CHARACTERISTICS OF THE POWER SOURCES

Power Source	Characteristics	
	Positive-sequence	Negative-sequence
MMC station	voltage source	/
DFIGs	current source	current source
PMSGs	current source	/
PVs	current source	/

B. Expressions of Post-fault Current

Different line faults can be analyzed as the combination of positive, negative, and zero-sequence networks. Take AG fault as an example, its composite sequence network is shown in Fig. 2, we represent positive and negative-sequence current sources at the RES side as equivalent sources I_{s+} , I_{dfig-} , impedance of the collector lines and transformers are represented as Z_{res+} , Z_{res-} , Z_{resT0} , impedance of the AC transmission lines at the RES side are represented as Z_{Lres+} , Z_{Lres-} , Z_{Lres0} . The same applies to the MMC side.

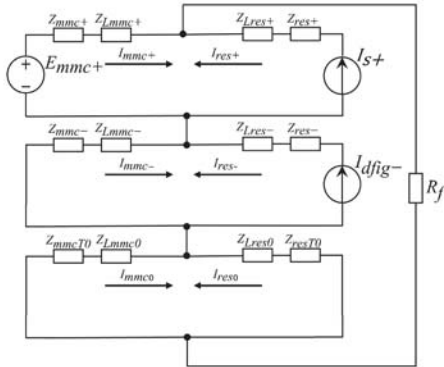


Fig. 2. Composite sequence network of AG fault.

Current at both ends can be described as (1), I_{res} represents the current at the RES side, and I_{mmc} represents the current at the MMC side.

$$\begin{cases} I_{mmc+} + I_{res+} = I_{mmc-} + I_{res-} = I_{mmc0} + I_{res0} \\ I_{res} = I_{res+} + I_{res-} + I_{res0} \\ I_{mmc} = I_{mmc+} + I_{mmc-} + I_{mmc0} \end{cases} \quad (1)$$

According to the superposition theorem, we have (2):

$$\begin{cases} I_{res+} = I_{s+} \\ I_{res-} = I_{dfig-} \\ I_{res0} = k_1 k_3 I_{s+} + k_2 k_3 I_{dfig-} + k_3 \frac{E_{mmc+}}{Z_{eq}} \\ I_{mmc+} = \frac{E_{mmc+}}{Z_{eq}} - (1 - k_1) I_{s+} + k_2 I_{dfig-} \\ I_{mmc-} = -\frac{E_{mmc+}}{Z_{eq}} + k_1 I_{s+} - (1 - k_2) I_{dfig-} \\ I_{mmc0} = (1 - k_3) \frac{E_{mmc+}}{Z_{eq}} + k_1 (1 - k_3) I_{s+} + k_2 (1 - k_3) I_{dfig-} \end{cases} \quad (2)$$

k_1 , k_2 are the distribution coefficients from source I_{s+} and I_{dfig-} to fault branch R_f , and k_3 is the distribution coefficient of the RES side in the zero-sequence network, which is $(Z_{Lmmc0} + Z_{mmcT0}) / (Z_{Lmmc0} + Z_{mmcT0} + Z_{Ls0} + Z_{sT0})$, k_{1-3} are complex numbers determined by the distribution of impedance in the composite sequence network. Z_{eq} is equivalent impedance, whose value is the summation of positive-sequence impedance, negative-sequence impedance, fault resistance, and zero-sequence impedance (zero-sequence of the RES side is paralleled with the MMC side). Time-domain expressions of the current at both ends are solved as (3), where ω^+ , ω^- are positive and negative-sequence angular frequency. $\theta_{\phi s+}$, $\theta_{\phi dfig-}$ and $\theta_{\phi mmc+}$ are the initial phase angles of the power sources.

$$\begin{cases} i_{res}(t) = (1 + k_1 k_3) i_{s+} \cos(\omega^+ t + \theta_{\phi s+}) \\ \quad + (1 + k_2 k_3) i_{dfig-} \cos(\omega^- t + \theta_{\phi s-}) \\ \quad + k_3 \frac{e_{mmc+}}{Z_{eq}} \cos(\omega^+ t + \theta_{\phi mmc+}) \\ i_{mmc}(t) = i_{s+} \cos(\omega^+ t + \theta_{\phi s+}) \\ \quad + i_{dfig-} \cos(\omega^- t + \theta_{\phi s-}) \\ \quad + (3 - k_3) \frac{e_{mmc+}}{Z_{eq}} \cos(\omega^+ t + \theta_{\phi mmc+}) \end{cases} \quad (3)$$

Current can be deduced similarly in the case of AB, ABG, and ABC faults. Complex coefficients and the combination of power sources will lead to different phase angles at both ends.

Considering the control strategies that restrict the current, post-fault characteristics at both ends of the AC transmission lines can be summarized as restricted amplitudes and controlled phase angles.

For traditional AC systems with current differential protection, phase angles of fault current at both ends are close, the differential current is significantly larger than the restraining current. However, when fault current amplitudes are restricted while a non-negligible difference between phase angles exists, the differential current may become smaller, current differential protection will face degradation in reliability. Fig. 3 shows the differential and restraining

current of phase *a* after an AB fault at the middle of the transmission line in our simulation model (fault resistance is 100Ω), differential current is smaller, causing failure of current differential protection. More such cases will be illustrated in Section IV.

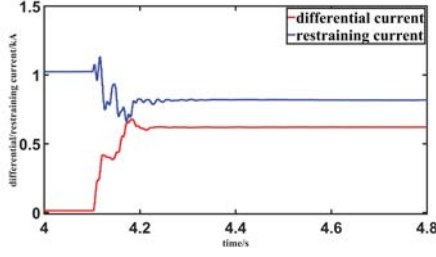


Fig. 3. Differential and restraining current after an internal AB fault.

III. TRAVELING WAVE PROTECTION SCHEME

A. Theory of Fault Traveling Waves

Voltage and current at any position of the line can be decomposed into forward traveling wave (FTW) and reverse traveling wave (RTW)[14]. FTW (u_f, i_f) and RTW (u_r, i_r) at a specific location can be deduced from instantaneous values of voltage and current as (4). x and t indicate the position and time of the traveling wave, v indicates the velocity of the traveling wave, and Z_C is the surge impedance.

$$\begin{cases} u_f \left(t - \frac{x}{v} \right) = \frac{u(x,t) + Z_C i(x,t)}{2} \\ u_r \left(t - \frac{x}{v} \right) = \frac{u(x,t) - Z_C i(x,t)}{2} \\ i_f \left(t - \frac{x}{v} \right) = \frac{u(x,t) + Z_C i(x,t)}{2Z_C} \\ i_r \left(t - \frac{x}{v} \right) = \frac{u(x,t) - Z_C i(x,t)}{2Z_C} \end{cases} \quad (4)$$

B. Decoupling of the Three-phase System

Considering the coupling of different phases in the AC transmission lines, a practical way for traveling wave analysis is to decouple the three-phase system into independent subsystems. We adopt Karenbauer transformation as (5), the three phases *a*, *b*, and *c* are transformed into modes 0, α , and β .

$$\begin{bmatrix} u_\alpha \\ u_\beta \\ u_0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} \quad \begin{bmatrix} u_\alpha \\ u_\beta \\ u_0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -2 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} \quad (5)$$

u_α, β are line-modes and u_0 is the zero-mode. For three-phase transmission lines, line-mode impedance is the same as positive-sequence impedance, and zero-mode impedance is the same as zero-sequence impedance. After line-mode and zero-mode impedance are calculated, surge impedance and velocity of the traveling wave in modal networks, together with FTW and RTW, can be calculated according to (4).

C. Traveling Wave Differential Protection

The traveling wave maintains a constant waveform while traveling through the distributed parameter transmission lines.

Considering the time delay, observation of the same traveling wave at both ends should be equal unless an internal fault occurs.

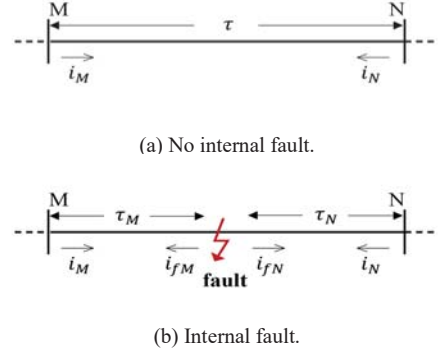


Fig. 4. Fault cases for traveling wave differential protection.

For the transmission line in Fig. 4, τ , τ_M , τ_N are time delays of the traveling wave on the total length of the line, on the interval between the fault and the M-end, and on the interval between the fault and the N-end. The direction of FTW is defined as pointing from the M-end to the N-end.

Based on the reference directions in Fig. 4, FTW and RTW at both ends can be calculated in phasor form as (6), frequency is 50Hz.

$$\begin{cases} \dot{I}_{Mf} = \frac{\dot{I}_M + \frac{\dot{U}_M}{Z_C}}{2} \\ \dot{I}_{Nf} = \frac{-\dot{I}_N + \frac{\dot{U}_N}{Z_C}}{2} \end{cases} \quad \begin{cases} \dot{I}_{Mr} = \frac{-\dot{I}_M + \frac{\dot{U}_M}{Z_C}}{2} \\ \dot{I}_{Nr} = \frac{\dot{I}_N + \frac{\dot{U}_N}{Z_C}}{2} \end{cases} \quad (6)$$

Select RTW differential current at the M-end and FTW differential current at the N-end as differential current. For protection units at both ends, the traveling wave differential current is calculated as (7):

$$\begin{cases} \dot{I}_{XM} = 2(\dot{I}_{Nr} e^{-j\omega\tau} - \dot{I}_{Mr}) \\ \dot{I}_{XN} = 2(\dot{I}_{Mf} e^{-j\omega\tau} - \dot{I}_{Nf}) \end{cases} \quad (7)$$

The proposed scheme is based on the integral value. We concentrate on the amplitude of the traveling wave differential current. The protection criterion is shown in (8):

$$\int_{t_0}^{t_0+t_w} |\dot{I}_X| dt > \varepsilon \quad (8)$$

t_0 is the starting instant of the time window, t_w is the width of the time window, and ε is the threshold value. The principle is shown in Fig. 5:

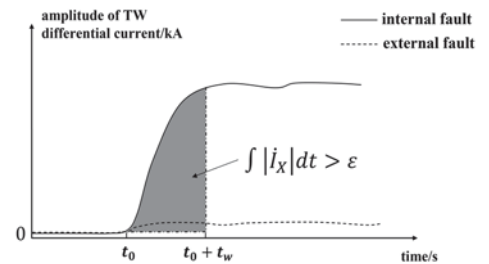


Fig. 5. Principle of traveling wave differential protection.

IV. SIMULATION ANALYSIS

A. Simulation Model

A simulation model of the system is built in PSCAD, the AC transmission line based on the Bergeron model is 30 km in length, and the voltage level is 220 kV. The bipolar structure is used on the MMC side. We use PMSG (which represents IIREG) and DFIG wind farm on the RES side. Detailed parameters are shown in TABLE II.

TABLE II. MODEL PARAMETERS

Models	Parameter	Value
DFIGs & PMSGs	rated active power of DFIG	$20 \times 5\text{MW}$
	rated active power of PMSG	$20 \times 5\text{MW}$
	Transformer wiring	Y-delta wiring with Y side (220kV) grounding
MMC station	Control strategy	u-f control
	rated voltage of the DC side	$\pm 320\text{kV}$
	Total capacity	200MVA
AC transmission line	Positive-sequence resistance	$0.1781598\text{e-}4\Omega/\text{m}$
	Positive-sequence inductive reactance	$0.31388\text{e-}3\Omega/\text{m}$
	Positive-sequence capacitive reactance	$273.5448\text{M}\Omega/\text{m}$
	Zero-sequence resistance	$0.2952201\text{e-}3\Omega/\text{m}$
	Zero-sequence inductive reactance	$0.1039898\text{e-}2\Omega/\text{m}$

Locations of faults are shown in Fig. 6. F1 and F5 are external faults, while F2-4 are internal faults. Different fault locations, fault types, and fault resistance are simulated in PSCAD. The sampling rate is set to 10 kHz.

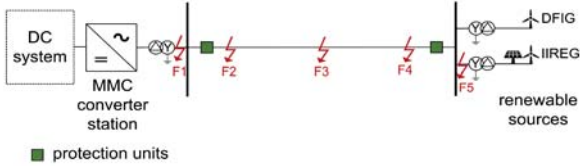


Fig. 6. Diagram of the simulation model and locations of faults.

B. Traveling Wave Differential Current

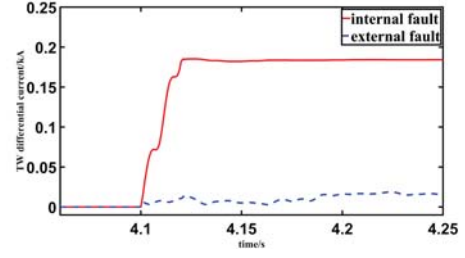
An ABG fault with a fault resistance of 220Ω is set at location F3, and an external fault (ABG metallic fault at location F1) is simulated. Calculate the 50Hz traveling wave differential current as follows:

1) *Basic parameters of traveling waves*: Calculate the surge impedance $Z_{Ca,0}$ and velocity $v_{a,0}$ of the traveling wave in the modal networks, and time delay $\tau_{a,0}$ based on positive and zero-sequence parameters of the transmission line.

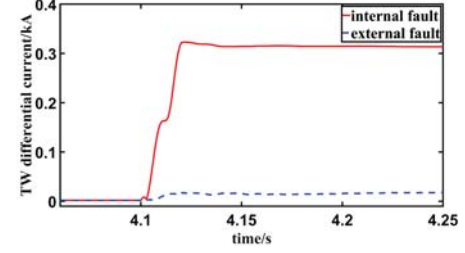
2) *Decoupling of three phases*: At the MMC and RES side, transform three-phase voltage and current into line-mode (α) and zero-mode voltage and current using Karenbauer transformation.

3) *Traveling wave differential current*: Calculate the FTW and RTW of current at both ends in modal networks based on surge impedance using (6), and calculate the traveling wave differential current using (7).

Traveling wave differential current in modal networks at both ends are calculated and shown in Fig. 7. Although the internal fault has a high fault resistance and the external fault is severe, the difference between them is significant.



(a) Zero-mode at the MMC side



(b) Line-mode at the RES side.

Fig. 7. Amplitude of the traveling wave differential current

C. Performance of Traveling Wave Differential Protection

Traveling wave differential protection introduced in Section III is tested, the width of the time window for integral (t_w) is 3 ms, line-mode and zero-mode thresholds are set to 2.598 and 1.139 to avoid false tripping under severest external faults. Fault recognition results and time (the time when the value of the integral exceeds the threshold) are recorded in TABLE III.

TABLE III. PERFORMANCE OF THE PROPOSED PROTECTION SCHEME

Fault Location & Type	Fault Resistance (Ω)	Recognition Time (ms)	
		Line-mode (MMC/RES side)	Zero-mode (MMC/RES side)
F1, ABC	0.001	Not trip/Not trip	Not trip/Not trip
	220	Not trip/Not trip	Not trip/Not trip
F2, AG	0.001	3.80/3.50	2.50/2.60
	220	15.8/11.3	8.30/8.40
F3, ABG	0.001	7.40/7.10	2.00/2.00
	220	8.10/7.80	3.90/3.90
F4, AB	0.001	5.80/5.60	Not trip/Not trip ^a
	220	6.70/6.60	Not trip/Not trip ^a
F5, AB	0.001	Not trip/Not trip	Not trip/Not trip
	220	Not trip/Not trip	Not trip/Not trip

^a. Zero-mode doesn't exist when the fault isn't grounded.

Simulation results show that the proposed protection scheme can reliably identify internal faults with high fault resistance and reliably prevent false tripping under external faults. Furthermore, fast fault recognition (within 10ms) can also be achieved with line-mode and zero-mode operating simultaneously.

Sensitivity also indicates the performance of relay protection. It is defined as the ratio between the action value and threshold value when the slightest internal fault occurs. We define the sensitivity of the proposed protection as the ratio between the integral value (of the steady-state post-fault differential current) and the threshold value. Several high fault resistance (220Ω) scenarios are recorded in TABLE IV. It is shown that the proposed traveling wave differential protection has high sensitivity.

TABLE IV. SENSITIVITY OF THE PROPOSED PROTECTION SCHEME

Fault Location	Fault Type	Sensitivity (line-mode/zero-mode)
F2	AG	1.908/4.840
	AB	7.273/0
	ABG	3.602/4.887
	ABC	2.261/0
F3	AG	1.903/4.820
	AB	7.212/0
	ABG	3.592/4.878
	ABC	2.424/0
F4	AG	1.905/4.773
	AB	7.152/0
	ABG	3.582/4.858
	ABC	2.556/0

D. Comparison with Current Differential Protection

As analyzed in Section I and II, when the phase angle is controlled, the differential current may become smaller than the restraining current, and current differential protection may fail to trip. This phenomenon is examined in our simulation model. We choose the vector summation of current at both ends as differential current and the vector subtraction of them as restraining current. TABLE V. lists some cases in which current differential protection may fail. Coefficient k is the ratio between the differential current and the restraining current, the typical value of k is 0.5-0.8, in the listed cases, k has minor values.

TABLE V. CASES CURRENT DIFFERENTIAL PROTECTION MAY FAIL

Fault Location	Fault Type	Fault Resistance (Ω)	k
F2	AB	100	0.754
F3	AB	100	0.759
	ABG	220	0.806
	ABC	100	0.806
	ABC	220	0.292
F4	AB	100	0.760
	ABG	220	0.733
	ABC	220	0.332

The analysis above shows that current differential protection will face degradation in reliability, fault resistance will worsen the impact. In contrast, the proposed traveling

wave differential protection correctly recognizes the faults, showing advantages in sensitivity and reliability.

V. CONCLUSION

In this paper, faults on AC transmission lines between MMC stations and RES are studied, characteristics of current amplitudes and phase angles are analyzed. Traveling wave protection solves the problem of sensitivity and reliability degradation of traditional protection schemes. By decomposing voltage and current into RTW and FTW, the protection scheme can concentrate on the essential feature of "whether the transmission channel of traveling waves has been destroyed by faults", thus successfully extracting the significant difference between internal and external faults.

A simulation model is built in PSCAD to verify the feasibility of the proposed traveling wave differential protection. Compared with current differential protection, traveling wave differential protection has advantages in reliability and the ability to withstand high fault resistance.

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